PIC Remaining Work: Workflows and Datasets

Basis	Product Blueprint /01 (flows F1–F9, §22 purchase types, §23 industry templates), Technical Document /02 (§6 state machines, §8 process definitions, §12 integration, §15 NFRs)
Evidence	Process definitions in `backend/app/pic/platform/process_defs.py`; step, branch and row counts from a full rehearsal of the Neon load (identical data to Neon)
Date	28 October 2026
Companion	`docs/PIC_Module_Gap_Analysis.md` (modules, screens, agents)

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1. Summary

Workflows. All nine blueprint flows are implemented: 84 process steps across F1–F9, each with its gateways, human tasks, SLAs and audit. The seeded data has run through **83 of the 84 steps**. The one step never reached is F4-10, contract renew / expire / terminate, because the seeded contracts are still within their validity.

What remains in the workflows is of three kinds:

Kind	Count	Examples
A. Branches that are built and tested but **never exercised by the seeded data**	22	Rejection and return paths, MRP simulation, GR reversal, ERS, invoice dispute, contract expiry and renewal, integration dead-letter and replay
B. **Functional pieces not built**	14	Punch-out, EDI/e-mail PO transmission, ERP planned-order import, reverse-auction live bidding, clarification board, contract termination and amendments, MDM-04 tax/address change, 4-way match
C. **External connections**	8	Real ERP posting, GST IRP, e-signature, bank payment files, sanctions/risk feeds, SSO, notifications, LLM

Datasets. Every PIC table that should have data does, with one exception: `fx_rate`, which waits for your rates. Two tables are empty on purpose:

- `reconciliation_item`: the platform and ERP reconcile with no differences;
- `idempotency_record`: it fills only from live API calls.

Three gaps remain in the datasets:

1. **Depth.** Only the Axion demo tenant exercises sourcing, contracts, receiving with quality, service entries, budgets, scorecards and forecasting. The migrated GMC and DBY tenants hold only what the legacy system had.
2. **Realism.** The Axion ERP history and the supplier twins are generated demo data. No real ERP extract has been loaded.
3. **Breadth.** There are no datasets for:
    - the five other industry templates;
    - the non-standard purchase types (expense, capital asset, consignment, subcontracting);
    - the volumes needed for the NFR tests.

2. Workflows F1–F9

Legend:

- **Exercised**: the seeded data passes through this branch.
- **Built, not exercised**: the code and tests cover it, but no seeded case does.
- **Not built**: the specification asks for it and it does not exist yet.

F1 · MRP run → planned order → requisition (`pic.mrp-to-pr`, 10 steps)

Exercised	Net-change run MR-1043 (AXN) and a regenerative run on the migrated stock (DBY), netting 12,000 − (1,400 + 1,200 + 600) + 200 = 9,000, planned order firmed and converted to REQ-2311 with source determination, BELOW_SAFETY exceptions, forecast override released to MRP
Built, not exercised	**Simulation run** (F1-05 "yes": results only, no documents); PAST_DUE and RESCHEDULE_IN/OUT exceptions; deletion of unfirmed planned orders by the next run; second plant P200; lot-sizing policies other than the seeded ones (EOQ, period, fixed)
Not built	**ERP planned-order import** (AD-05, for tenants that run MRP in the ERP); factory calendar maintenance screen; WBS and reservation demand

F2 · Guided buying → requisition (`pic.requisition`, 9 steps)

Exercised	Contract, catalogue, material and free-text routes (F2-03); reusable item → parallel MDM request (F2-05 yes and no); budget check; DMN-01 approval chains, including the auto-approved catalogue line under ₹25,000
Built, not exercised	**Approver rejects or returns** a requisition (all 42 seeded task decisions are approvals); budget **hard stop** for capex/WBS; soft stop with justification; capex committee for asset lines; re-approval after a value, account or supplier change; DMN-01 band edges (24,999 / 25,000 / 2 L / 10 L)
Not built	**Punch-out** (cXML / OCI) carts returning as catalogue lines; P-card policy; mobile approval UI

F3 · Strategic sourcing → award (`pic.sourcing-award`, 10 steps)

Exercised	RFQ-2231 with weights locked at publish, sealed bids, TCO and should-cost, a second negotiation round (F3-05 yes), capacity-constrained award scenarios, 60/40 award above ₹50 L with approval and conflict declaration; legacy RFQ-2026-092 (two rounds)
Built, not exercised	Award below ₹50 L and lowest bid (F3-07 "no", no approval needed); event cancellation; early-unseal attempt being blocked (tested in the suite, not seeded)
Not built	**Clarification Q&A board** (F3-03 "clarifications" exists only as the step name); **reverse-auction live bidding**; RFI/RFP-specific templates (the event types exist; RFQ is the only one seeded)

F4 · Contract lifecycle (`pic.contract`, 10 steps)

Exercised	Draft from award with playbook clauses, AI redline, supplier redline, approval above ₹1 Cr (F4-05 yes), two-party signature, activation publishing prices and the 60/40 quota, LME index reset creating new validity periods, consumption tracking; a manual value contract drafted (SA-5522)
Built, not exercised	**F4-10 renewal / expiry**: the renewal timer fires 120 days before expiry and the expiry timer at validity end, but no seeded contract has reached either; approval below ₹1 Cr (F4-05 no); consumption threshold alerts at 80/90/100% (no contract has reached 80%); completing an obligation
Not built	**Termination** and **amendments / variations** as contract versions; external **e-signature** provider (signatures are recorded internally with a document hash); AI extraction of clauses from uploaded legacy contracts

F5 · Purchase order → confirmation → delivery (`pic.po`, 10 steps)

Exercised	Inheritance from requisition, contract and source list; risk score; touchless release (F5-04 yes) and every non-touchless reason (info record, manual, quote); approval; confirmation within tolerance; buyer change order (v2); supplier change request after confirmation (v3); expediting follow-up
Built, not exercised	**Supplier proposes a change outside tolerance at confirmation** (F5-08 "no" → buyer review); **supplier rejects** the PO; confirmation SLA breach → expediting task; release to a blocked supplier being refused; PO cancellation and **closure** (no PO is CLOSED)
Not built	**Transmission by EDI (ORDERS/ORDCHG/ORDRSP) or e-mail** (portal only); schedule lines (multiple delivery dates per line)

F6 · Goods receipt / service entry → quality (`pic.receipt`, 9 steps)

Exercised	GR against ASN; quality-managed lot GR-90231 with 5,380 accepted and 20 rejected → return delivery RD-4410 → CAPA-301; non-quality GR; service entry sheets approved; stock, GR/IR and scorecard updates
Built, not exercised	**GR reversal**; fully rejected lot; over-delivery above tolerance being blocked; **consignment** GR (liability on withdrawal) and **subcontracting** GR (component consumption), configured in the purchase-type matrix but no PO of those types; SES rejection
Not built	**E-way bill** capture on ASN; **EDI DESADV**; batch/serial/expiry traceability beyond the batch field; measurement- or milestone-based SES (construction / IT templates)

F7 · Invoice → match → ERP posting → payment (`pic.invoice`, 9 steps)

Exercised	Portal capture, hard and fuzzy duplicate check, 2-way, 3-way and SES match, quantity exception → credit note, price exception → variance accepted, duplicate rejected, idempotent posting with replay, payment runs, realised savings; legacy capture with 11 duplicates and 3 missing-GR holds; reconciliation REC-101
Built, not exercised	**ERS** (self-billing) for consignment and scheduling agreements; **dispute** resolution; tax exception (IRN mismatch); **ERP timeout, retry, dead-letter and replay** (covered by tests with simulated faults, not in the seed); bank-change cooling-off blocking a payment

Not built	**4-way match** (named in F7-03); **GST IRP** validation of IRN and QR; **bank payment file** (H2H / NEFT batch); EDI INVOIC; OCR capture of PDF invoices (the channel exists, extraction does not)

F8 · Supplier onboarding & qualification (`pic.supplier-onboarding`, 9 steps)

Exercised	Registration, profile and certificates, screening (duplicates and sanctions: ZAT006 blocked), dual bank verification with call-back, qualification QUALIFIED and CONDITIONAL with a waiver, ERP sync, risk monitoring
Built, not exercised	**Qualification rejected** (F8-06 "no"); **certificate expiry** turning a supplier Conditional (all 8 certificates are valid); Preferred / Exited status changes; block with open POs
Not built	Supplier **self-registration by invitation e-mail**; questionnaire builder; licensed sanctions / financial / news feeds

F9 · Material master governance (`pic.material-request`, 8 steps)

Exercised	Duplicate score below 0.92 → four domain stewards → ERP sync → linked to the PR line (MRQ-7731); duplicate ≥ 0.92 merged (MRQ-7732); a migrated request in stewardship (DBY)
Built, not exercised	**Steward rejects** a request; steward override of a duplicate block with a reason; the 0.75–0.92 warning band
Not built	**MDM-04 supplier tax-ID and address changes** (bank changes are built); bulk material import / cleansing

Process step coverage

Flow	Steps	Executed	Gateways exercised on both branches
F1	10	10	0 of 1 (simulation never run)
F2	9	9	2 of 2
F3	10	10	1 of 2
F4	10	9	0 of 1
F5	10	10	1 of 2
F6	9	9	2 of 2
F7	9	9	1 of 1
F8	9	9	1 of 1 (qualified and conditional; rejected never)

Flow	Steps	Executed	Gateways exercised on both branches
F9	8	8	1 of 1
Total	**84**	**83**	**9 of 13**

3. Workflows of the intelligence and governance modules

Module flow	Exercised	Remaining
Spend classification → steward review	218 auto-classified, 48 reviewed (44 decided, 4 open); legacy lines queued	**Reclassification** by a steward (all decisions so far confirmed the proposal); a UNSPSC or customer taxonomy import
Maverick hit → disposition	446 hits in total; policy breach and recovery dispositions	Accepted-exception and false-positive dispositions; recovery follow-up into a debit note
Category plan → approval	15 plans, ALU-DIECAST approved	Plan budget used as a **budget check** at requisition (category budgets are not yet linked to DMN-01)
Forecast → Demand Sensing → release	DC-331 over-forecast by 22% → recommendation accepted → override released	A **rejected** recommendation; forecasts for legacy materials (too few goods issues)
Risk signal → mitigation	8 signals, recommendations raised	**Mitigate / close** a signal; feed ingestion (adapter ready, no feed)
CAPA	CAPA-301 opened from a rejection and responded to by the supplier	**Quality verification and closure** (left open on purpose in the seed)
Reconciliation	REC-101, 0 differences	A difference and its resolution (needs a real ERP or a simulated mismatch)
Agent autonomy	Defaults and on-demand runs	Automatic **suspension** when acceptance falls below 70% (needs ≥ 20 feedback events)

4. Datasets

4.1 Coverage by tenant

Area	AXN (demo, manufacturing)	GMC (migrated)	DBY (migrated)
Organisation	1 company code, 2 plants, 2 storage locations, 3 cost objects	1 / 1 / 1 / 2	1 / 1 / 4 / 2
Users	31	30	24
Suppliers	26 (6 onboarded through F8 and 20 ERP vendor masters)	5	1
Materials	4	0	9
Requisitions / POs	7 / 9	5 / 5	7 / 6
Sourcing / contracts	1 event, 3 contracts	—	1 event, —
Receiving	5 ASN, 4 GR, 3 inspection lots, 2 SES	—	3 ASN, 3 GR
Invoices / payments	6 / 4	5 / 2	24 / 13
Spend lines	2,573 (24 months ERP history + platform)	2	13
Forecasts / MRP	60 forecast periods, 1 MRP run	—	1 MRP run
Scorecards, certificates, twin	5 / 8 / 4	—	—
Budgets, savings	3 budgets, 2 initiatives	—	—

GMC and DBY are thin because the legacy system held little: 156 rows in 15 tables, no contracts, no certificates, no budgets. They can only become richer from real ERP data (§4.4).

4.2 Empty tables

Table	Why it is empty	What fills it
`fx_rate`	No rates were invented	You provide treasury rates (ADM-01 or the API); needed if the legacy amounts are USD
`reconciliation_item`	No differences between platform and ERP	A real ERP, or a seeded mismatch case (§5 A)
`idempotency_record`	Filled only by API calls with an Idempotency-Key	Live use

4.3 Generated or placeholder data

The data in this table is generated or a placeholder; it must be replaced before production.

Data	Status	Replace with
AXN ERP spend history (2,568 lines, Oct 2024 – Sep 2026) and 20 ERP vendors	Generated, calibrated to blueprint AIX-01	A real SAP/ERP extract: 24–36 months of invoices and POs (TD §12.4)
AXN consumption history (DC-330, DC-331, FK-M8-50)	Generated	The ERP stock and consumption feed
Supplier twins (capacity, trust, late probability)	Seeded for 4 suppliers	The Supplier Capability Digital Twin service
Sanctions list	4 fictitious SAMPLE-LIST names	A licensed sanctions feed
Certificates	8 records, all valid	Supplier uploads with expiry dates
Bank accounts of migrated suppliers	Inactive placeholders	MDM-04 bank changes by the suppliers
GST / IRN	IRN computed locally	GST IRP validation
Industry templates	Configuration for 6 packs; data only for manufacturing	The demo datasets in §4.4

4.4 Datasets not created yet

#	Dataset	Spec	Purpose	Size
D1	**Branch cases** for the 22 unexercised branches in §2, seeded as operating cases 9–n so every gateway has both outcomes	TD §17 E2E	Demo and regression coverage	M
D2	**Purchase-type cases**: expense (receipt-less below threshold), capital asset (capex committee + WBS hard stop), consignment (GR to consignment stock, ERS settlement), subcontracting (component consumption)	BP §22	P08 differentiation	M
D3	**Construction** demo tenant: WBS C-2207 "Tower B structure", subcontract SC-118, SES at 35% progress, 5% retention, variation order	BP §23	Industry pack	L
D4	**IT services** demo tenant: SOW-4410 T&M, rate card RC-22, timesheet SES 320 h, licence renewal alert	BP §23	Industry pack	M
D5	**Logistics** demo tenant: lane CHE→PNQ 32 ft MXL, rate 54/km + fuel index, POD confirmation, freight audit	BP §23	Industry pack	M
D6	**Healthcare** demo tenant: consignment implant KN-77, lot/expiry, usage-triggered PO, recall trace	BP §23	Industry pack	L
D7	**General enterprise** demo tenant: laptop punch-out order ₹78,500, single approval	BP §23	Industry pack (needs punch-out)	M
D8	**Volume datasets** for NFRs: 50,000 materials (MRP < 10 min), 5 M invoices/year, 50 M spend lines (cube < 3 s), 10,000 concurrent users	TD §15	Performance tests	L
D9	**AI evaluation sets**: labelled spend lines (classification accuracy), forecast actuals (MAPE), duplicate pairs (dedup precision/recall), Copilot questions (answer-contract completeness ≥ 99%)	TD §17	Before raising autonomy levels	M

#	Dataset	Spec	Purpose	Size
D10	**Real tenant masters**: materials, suppliers (with bank and tax IDs), contracts, open POs and open AP from your ERP	TD §12.4	Pilot	Depends on ERP
D11	**Reference data**: FX rates, HSN/SAC and GST rates, holiday calendars per plant, UNSPSC or your own taxonomy	TD §4, BP §7	Correct tax, dates and classification	S

Sizes: S ≈ 1–2 days, M ≈ 3–5 days, L ≈ 1–2 weeks of build and verification.

5. Work plan

A. Can be built now (no external input)

Priority	Item	Covers
1	Operating cases 9–20 for the unexercised branches. Each needs its own seeded case: requisition rejected and returned; MRP simulation; PAST_DUE / reschedule exceptions; GR reversal; fully rejected lot; supplier rejects a PO; change outside tolerance; ERS settlement; invoice dispute; ERP timeout → dead-letter → replay; certificate expiry → Conditional; qualification rejected; steward rejects a material; contract below ₹1 Cr; contract reaching 80/90/100%; contract expiry → renewal decision; PO closure	§2 A, D1
2	Purchase-type cases: expense, capital asset with capex committee and WBS hard stop, consignment, subcontracting	D2
3	Contract termination and amendments (variation orders as versions)	F4
4	MDM-04 tax-ID and address changes with dual approval	F9 / M03
5	Clarification board and reverse-auction live bidding	F3
6	4-way match (PO + GR + inspection + invoice) and the category budget check at requisition	F7, M01
7	ERP planned-order import adapter (file / API)	F1, AD-05
8	Volume generator and the k6 / MRP / cube performance runs (on a production-like environment)	D8
9	AI evaluation sets from the seeded and migrated data, plus a labelling screen for stewards	D9
10	Industry demo tenants D4 and D5 (no external system needed), then D3 and D6	Phase 4

B. Needs your input or a third-party system

Real ERP posting and inbound masters (D10), GST IRP, e-signature, bank payment files, EDI / punch-out (needed for D7), sanctions and risk feeds, SSO / MFA, e-mail/SMS notifications, the LLM gateway.

6. What you need to supply

1. **An ERP extract or sandbox** (SAP S/4HANA first): vendors, materials, contracts, open POs, open AP and 24–36 months of invoice/PO history. This replaces the generated AXN history and fills GMC/DBY properly.
2. **FX rates** and the currency of the legacy amounts.
3. **Reference data**: HSN/SAC master and GST rates, plant holiday calendars, your spend taxonomy (or confirm UNSPSC).
4. **Credentials**: GST IRP / GSP, e-sign provider, bank host-to-host, EDI / punch-out suppliers, sanctions feed, identity provider, LLM gateway.
5. **The test workbook** AHG/PRC/PIC/2026-10/03, so that the branch cases (§5 A1) match its E2E numbering.
6. **Priorities**: which purchase types and which industry template come first.